

Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d +Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

Principal authors: Jana Bielcikova, Christine Nattrass, Helen Caines, Betty Abelev, Rene Bellwied, Marek Bombara, Leon Gaillard
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We present results on the near-side of high- p_T triggered di-hadron correlations of neutral strange baryons (Λ , $\bar{\Lambda}$) and mesons (K_S^0) at intermediate $p_T = 2$ -6 GeV/ c in d +Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The near-side di-hadron correlation in $(\Delta\phi, \Delta\eta)$ contains two structures, a peak which is narrow in $\Delta\phi$ and $\Delta\eta$ consistent with correlations due to jet fragmentation and correlation in $\Delta\phi$ which is broad in $\Delta\eta$ which is attributed dominantly to flow. The dependence of the conditional yield of the jet-like correlation on the trigger particle momentum, associated particle momentum, and centrality are presented. In addition the particle composition of the jet-like correlation is determined using identified associated particles. Data on identified triggers are qualitatively described by PYTHIA, however the composition of the jet-like correlation is not quantitatively described by PYTHIA.

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I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Measurements of nuclear modification factors [1–4] have shown that the nuclear medium created is opaque to particles with large transverse momentum (p_T). Moreover, anisotropic elliptic flow measurements demonstrate that the medium exhibits partonic degrees of freedom and has properties close to those expected of a perfect fluid [5, 6].

Studies of jets in heavy ion collisions are possible through single particle measurements [1–4], di-hadron correlations [7–17], and measurements of reconstructed jets [3, 18–21]. Measurements of reconstructed jets are useful for studying higher momenta and have provided direct evidence for partonic energy loss in the medium. However, di-hadron correlations provide complementary measurements since they enable studies at intermediate momenta, where the interplay between jets and the medium is important and direct jet reconstruction is no longer possible.

Properties of jets have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum at the Relativistic Heavy Ion Collider (RHIC) [7–14] and the Large Hadron Collider (LHC) [15–17]. Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed significant modification, including the disappearance of the peak at intermediate p_T (< 2 GeV/ c) [10, 22] and its reappearance at high p_T [11, 23]. The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central Au+Au collisions [8, 12, 24]. In p + p and d +Au collisions, there is a peak that is narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to as the jet-like correlation. In Cu+Cu and Au+Au col-

lisions this peak is observed to be broader than that in d +Au collisions, although the yields are comparable [7]. This indicates that the jet-like correlation is dominantly produced through fragmentation.

Besides the shape modifications of jet-like correlations at intermediate transverse momenta, the production mechanism of hadrons may differ from simple fragmentation. In central Au+Au collisions baryon production in the bulk is enhanced relative to that in p + p or d +Au collisions [25, 26]. The baryon to meson ratios measured in Au+Au collisions increase with p_T until reaching a maximum of ≈ 3 relative to p + p collisions at $p_T \approx 3$ GeV/ c in both the strange and non-strange quark sectors. A fall-off of the baryon to meson ratio is observed for $p_T > 3$ GeV/ c and both the strange and non-strange baryon to meson ratios in Au+Au collisions approach the values measured in p + p collisions at $p_T \approx 6$ GeV/ c .

In this paper, studies of two-particle correlations on the near-side in d +Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment are presented. Results from two-particle correlations in pseudo-rapidity and azimuth for neutral strange baryons (Λ , $\bar{\Lambda}$) and mesons (K_S^0) at intermediate p_T ($2 < p_T < 6$ GeV/ c) in the different collision systems are compared to unidentified charged hadron correlations. Both identified trigger particles associated with charged particles (K_S^0 -h, Λ -h) and unidentified charged trigger particles associated with identified strange particles (h- K_S^0 , h- Λ) are studied. The near-side jet-like yield is studied as a function of centrality of the collision and transverse momentum of trigger and associated particles to look for possible flavor and baryon/meson dependence. The composition of the jet-like correlation is studied using identified associated particles to investigate possible production mechanisms. The results are compared to expectations from PYTHIA.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

STAR [27] is a multi-purpose spectrometer with a full azimuthal coverage consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. This analysis is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [28] with a pseudo-rapidity acceptance $|\eta| < 1.5$. The TPC has in total 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements for each charged particle track. The TPC momentum resolution is determined to be $\Delta p/p \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [28]. Charged particle tracks used in this analysis were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm and a pseudorapidity $|\eta| < 1.0$. The results presented in this paper are based on analysis of data from d +Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV taken by the STAR experiment in 2003, 2005, and 2004, respectively.

For d +Au collisions, the events analyzed were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located 18 m from the nominal interaction point in the Au beam direction and accepting $95 \pm 3\%$ of the hadronic cross section [29]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) placed at forward pseudorapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the two ZDCs. The MB Au+Au events required a coincidence between the two ZDCs, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats aligned parallel to the beam axis and arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions. An additional online trigger for central Au+Au collisions was used to sample the most central 12% of the total hadronic cross section. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB.

In order to achieve a more uniform detector acceptance in Cu+Cu and Au+Au data sets, only those events with the primary vertex position along the longitudinal beam direction (z) within 30 cm of the center of the STAR detector were used for the analysis. For the d +Au collisions this vertex position selection was extended to $|z| < 50$ cm. The number of events after the vertex cuts in individual data samples is summarized in Table I.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	No. of events [10^6]
d +Au	0-95%	3
Cu+Cu	0-60%	38
Au+Au	0-80%	28
Au+Au	0-12%	17

We identify weakly decaying neutral strange (V^0) particles Λ , $\bar{\Lambda}$ and K_S^0 by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC as described in [30]:

$$\begin{aligned}\Lambda &\rightarrow p + \pi^-, BR = (63.9 \pm 0.5)\% \\ \bar{\Lambda} &\rightarrow \bar{p} + \pi^+, BR = (63.9 \pm 0.5)\% \\ K_S^0 &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\%\end{aligned}\quad (1)$$

where BR denotes the branching ratio. The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates. Reconstructed Λ and K_S^0 particles are required to be within $|\eta| < 1.0$. Topological cuts are optimized for each data set and chosen to reach a signal to background ratio of at least 15:1. For the analyses presented here, no difference was observed between results with Λ and $\bar{\Lambda}$ trigger particles. Therefore the correlations were added to increase the statistical significance of the results. In the remainder of the discussion the combined particles are referred to simply as Λ baryons.

III. METHOD

A. Correlation technique

The analysis in this paper follows the method in [7]. A high- p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger particle in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) is formed. This distribution, $d^2 N_{\text{raw}}/d\Delta\phi d\Delta\eta$, is normalized by the number of trigger particles, N_{trigger} , and corrected for the efficiency and acceptance of associated tracks:

$$\frac{d^2 N}{d\Delta\phi d\Delta\eta}(\Delta\phi, \Delta\eta) = \frac{1}{N_{\text{trigger}}} \frac{d^2 N_{\text{raw}}}{d\Delta\phi d\Delta\eta} \frac{1}{\varepsilon_{\text{assoc}}(\phi, \eta)} \frac{1}{\varepsilon_{\text{pair}}(\Delta\phi, \Delta\eta)}. \quad (2)$$

The efficiency correction $\varepsilon_{\text{assoc}}(\phi, \eta)$ is a correction for the single particle reconstruction efficiency in TPC and $\varepsilon_{\text{pair}}(\Delta\phi, \Delta\eta)$ is a correction for the finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$. Since the correlations are normalized by the number of trigger particles, the efficiency correction is only applied for the associated particle. The fully corrected correlation functions are averaged between positive and negative $\Delta\phi$ and $\Delta\eta$ regions and are reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$ in the plots.

B. Single charged track efficiency correction

For unidentified charged associated particles, the efficiency correction $\varepsilon_{\text{assoc}}(\phi, \eta)$ is the correction for charged hadrons, identical to that applied in [7]. This single charged track reconstruction efficiency is determined as a function of p_T , η , and centrality by simulating the TPC

response to a particle and embedding the simulated sig-
nals into a real event. The single charged track efficiency
is found to be approximately constant for $p_T > 2$ GeV/ c
and ranges from around 75% for central Au+Au events to
around 85% for peripheral Cu+Cu events. The efficiency
for reconstructing a track in d +Au events is 89%.

For identified associated strange particles, the recon-
struction efficiency $\varepsilon_{\text{assoc}}(\phi, \eta)$ is determined in a similar
way, but forcing the simulated particle to decay through
the channel in Equation 2 and then correcting for the
respective branching ratio. The efficiency for Λ , $\bar{\Lambda}$, and
 K_S^0 ranges from 8% to 15%, increasing with momentum
and decreasing with system size [31]. No correction for
the reconstruction efficiency is applied for identified trig-
ger particles because the reconstruction efficiency does
not vary significantly within the p_T^{trigger} bins used in this
analysis.

The systematic uncertainty associated with the effi-
ciency correction for unidentified associated hadrons, 5%,
is strongly correlated across centralities and p_T bins for
each data set but not between data sets. For identified
associated particle ratios the systematic errors on the ef-
ficiency correction partially cancel out and are negligible
compared to the statistical uncertainties.

For the inclusive spectra the feeddown correction due
to secondary Λ baryons from Ξ baryon decays is 15%,
independent of p_T [32]. For identified Λ trigger particles,
we assume that feeddown lambdas do not change the
correlation since little trigger particle type dependence is
observed in the data. This is consistent with the available
data within errors. For identified associated particles,
we assume the same correlation between primary and
secondary Λ baryons and correct the yield of Λ associated
particles by reducing the yield by 15%.

C. Pair acceptance correction

The requirement that each track falls within $|\eta| < 1.0$
in TPC results in a limited acceptance for track pairs.
The geometric acceptance for a track pair is $\approx 100\%$
for $\Delta\eta \approx 0$ and close to 0% near $\Delta\eta \approx 2$. The track
pair acceptance is limited in azimuth by the 12 TPC
sector boundaries, leading to dips in the acceptance of
track pairs in $\Delta\phi$. To correct for the limited geometric
acceptance, a mixed event analysis was performed using
trigger particles from one event combined with associated
particles from another event, as done in [12]. The event
vertices were required to be within 2 cm of each other
along the beam axis and the events were required to have
the same charged particle multiplicity within 10 particles.
To increase statistics of the mixed event sample, each
event with a trigger particle was mixed with ten other
events.

D. Track merging correction

The track merging effect in unidentified hadron (h)
correlations discussed in [7] is also present for V^0 -h and
h- V^0 correlations. This effect leads to a dip at small $\Delta\phi$
and $\Delta\eta$ due to overlap between the trigger and associ-
ated particles. When one of the particles is a V^0 , this
overlap is between one of the V^0 daughter particles and
the unidentified hadron. The size of the dip due to track
merging depends strongly on the relative momenta of the
particle pair. This effect is larger when the overlapping
particles' momenta are closer. For V^0 -h correlations, the
typical associated particle momentum is approximately
1.5 GeV/ c . Since the K_S^0 decay is symmetric, this means
that the track merging effect is greatest for K_S^0 -h cor-
relations with a trigger K_S^0 momentum of approximately 3
GeV/ c . In a Λ decay, the proton daughter carries more
of the Λ momentum than the pion daughter. Therefore
this effect is larger for Λ trigger particles with lower mo-
menta. Because track merging affects both signal and
background particles and the signal sits on top of a large
combinatorial background, the effect is larger for colli-
sions with a higher charged track multiplicity. Since the
dip in V^0 -h and h- V^0 correlations is the result of a V^0
daughter merged with an unidentified hadron, the dip is
wider in $\Delta\phi$ and $\Delta\eta$ than in unidentified hadron cor-
relations. When the track merging dip is present, it is cor-
rected by fitting a Gaussian to the peak, excluding the
region impacted by track merging, and using the Gaus-
sian fit to extract the yield.

In the absence of a clear dip from track merging, the
yield calculated from bin counting is consistent with the
yield from the Gaussian fit. When the residual track
merging effect is corrected by a fit, the track merging
correction applied by the fit is approximately the same
order as the statistical error on the yield. We therefore
conclude that when no dip is evident, the track merging
effect is negligible compared to the statistical error on
the yield.

E. Yield extraction

The notation and method used to extract the yield in
this paper follow [7, 12]. The jet-like correlation is narrow
in both $\Delta\phi$ and $\Delta\eta$ and is contained within the kinematic
cuts in p_T^{trigger} and $p_T^{\text{associated}}$ used in this analysis. The
di-hadron correlation from Equation 2 is projected onto
the $\Delta\eta$ axis:

$$\left. \frac{dN}{d\Delta\eta} \right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2 N}{d\Delta\phi d\Delta\eta}. \quad (3)$$

All other correlations, including those from v_2 , v_3 , and
higher order flow harmonics, are assumed to be indepen-
dent of $\Delta\eta$ within the η acceptance of the analysis, con-
sistent with [12, 33–35]. We make the assumption that

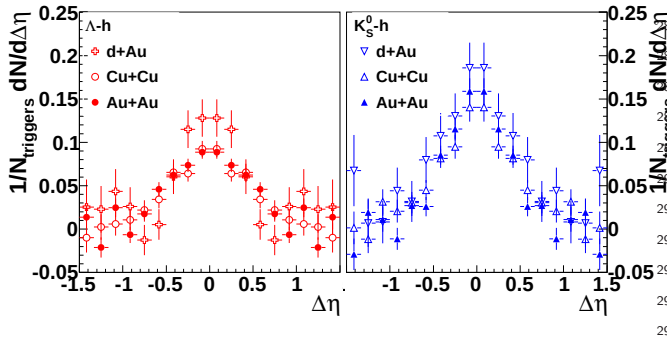


FIG. 1: Corrected correlation functions $\frac{dN_J}{d\Delta\eta}$ in $|\Delta\phi| = 1.0$ for $3 < p_T^{\text{trigger}} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\text{associated}} < p_T^{\text{trigger}}$ for Λ -h (left) and K_S^0 -h (right) for minimum bias d +Au, 0-20% Cu+Cu, and 40-80% Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV after background subtraction. The data have been reflected about $\Delta\eta = 0$.

the η dependence observed for v_3 measured using the two particle cumulant method [36] is entirely due to nonflow. This method explicitly subtracts such correlated backgrounds in addition to uncorrelated background. This means that the jet-like yield can be determined by:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \frac{dN}{d\Delta\eta} \Big|_{-0.78, 0.78} - b_{\Delta\eta} \quad (4)$$

where $b_{\Delta\eta}$ is a constant offset determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. Variations in the method for extracting the constant background, such as fitting a constant at large $\Delta\eta$, lead to differences in the yield smaller than the statistical error due to the background alone. Sample correlations are given in Figure 1. When the dip due to track merging is negligible, the yield determined from fit is discarded to avoid any assumptions about the shape of the peak and instead we integrate Equation 4 over $\Delta\eta$ using bin counting to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \frac{dN_J}{d\Delta\eta}(\Delta\eta). \quad (5)$$

Where track merging is negligible the yield from the fit and from bin counting are consistent. Therefore for data where the dip due to track merging is significant, the yield from the Gaussian fit is used, excluding the region affected by track merging, and the statistical errors from the fit exceed any error from the assumption of a Gaussian shape.

IV. RESULTS

A. Charged- V^0 correlations

Previous studies demonstrated that the jet-like correlation in h-h correlations is independent of collision system [7, 12, 37] and that it is qualitatively described by

PYTHIA [7]. This indicates that the jet-like correlation is dominantly produced by fragmentation, even at intermediate momenta ($2 < p_T < 6$ GeV/c) where recombination is expected to contribute substantially to the bulk production of particles. The composition of the jet-like correlation can be studied using correlations with identified associated particles. For the analysis presented here, the statistics of the d +Au data were limited and the Au+Au data set was limited by the presence of residual track merging. Therefore it was only possible to determine the composition of the jet-like correlation in Cu+Cu collisions.

These measurements are compared to inclusive baryon to meson ratios in p + p collisions from STAR [38] and ALICE [39] and simulations in p + p collisions in PYTHIA [40] using the Perugia 2011 [41] tune and Tune A [42] in Figure 2. The ratio in Cu+Cu collisions is consistent with the inclusive particle ratios from p + p . This further supports earlier observations that the jet-like correlation is dominantly produced by fragmentation since the production of particles both in p + p collisions at these momenta and in the jet-like correlation are expected to be dominated by fragmentation. This implies that production of strange particles through recombination is not significant in the jet-like correlation, even in A+A collisions where the inclusive spectra show an enhancement of Λ of up to a factor of three relative to the K_S^0 .

The inclusive particle ratios at $\sqrt{s} = 200$ and 7000 GeV are consistent with each other, however, the particle ratios are not described by PYTHIA. This is because PYTHIA has difficulty with both strangeness and baryon production. PYTHIA is able to describe π and ω production [43, 44], however, generally underestimates production of strange particles, including the K_S^0 , and baryons, including the Λ [38, 39, 43, 44]. Tune A has been tuned to match low momentum h-h correlations [42] while the Perugia 2011 tune has been chosen to match inclusive particle spectra better, including recent data from the LHC [41].

PENDING The h- K_S^0 and h- Λ correlations are likely dominated by gluon and light quark jets, indicating that PYTHIA is underestimating the production of strange particles in light quark jets. However, K_S^0 -h and Λ -h correlations should have larger contributions from jets containing strange particles. It may be easier for PYTHIA to describe the fragmentation of an initial strange quark than to describe the production of strange particles from non-strange jets.

B. Correlations with identified strange trigger particles

The jet-like yield as a function of p_T^{trigger} is shown in Figure 3 for K_S^0 -h, and Λ -h correlations for d +Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data are tabulated in Table II. Due to residual track merging effects, fits are used for Λ -h correlations

TABLE II: The jet-like yield in $|\Delta\eta| < 0.78$ as a function of p_T^{trigger} for K_S^0 -h, and Λ -h correlations for $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$ in minimum bias d +Au, 0-60% Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$, as shown in Figure 3.

p_T	K_S^0 -h				Λ -h			
	MB d +Au	0-60% Cu+Cu	40-80% Au+Au	0-12% Au+Au	MB d +Au	0-60% Cu+Cu	40-80% Au+Au	0-12% Au+Au
2.0-2.5	—	0.036 ± 0.004	0.063 ± 0.008	—	—	0.026 ± 0.005	—	—
2.5-3.0	—	0.059 ± 0.006	0.084 ± 0.023	—	—	0.071 ± 0.007	0.061 ± 0.010	—
3.0-3.5		0.098 ± 0.009	0.139 ± 0.022	—		0.084 ± 0.017	0.100 ± 0.019	0.105 ± 0.022
3.5-4.0	0.162 ± 0.028		0.172 ± 0.021	0.160 ± 0.036	0.079 ± 0.018		0.096 ± 0.030	0.128 ± 0.022
4.0-4.5		0.144 ± 0.011				0.142 ± 0.013		
4.5-5.0			0.170 ± 0.037	0.240 ± 0.045			0.184 ± 0.040	0.091 ± 0.033
5.0-5.5	—	—			—	—		

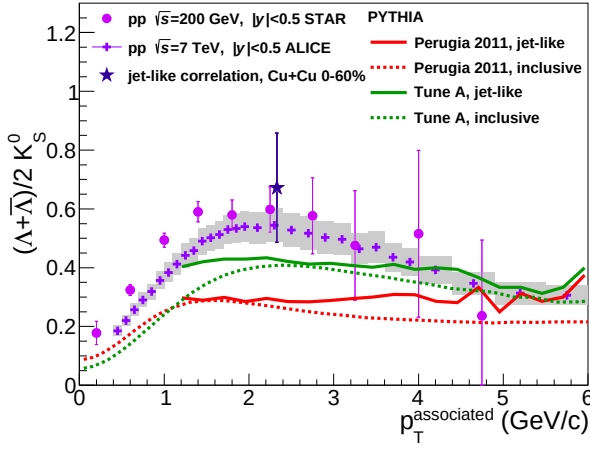


FIG. 2: Λ/K_S^0 ratio measured in the jet-like correlation in 0-60% Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV for $3 < p_T^{\text{trigger}} < 6$ GeV/c and $2.0 < p_T^{\text{associated}} < 3.0$ GeV/c along with this ratio obtained from inclusive p_T spectra in p+p collisions. Data are compared to calculations from PYTHIA [40] using the Perugia 2011 tunes [41] and Tune A [42]. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

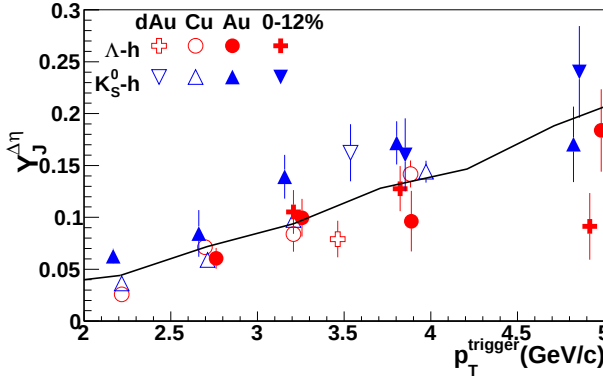


FIG. 3: The jet-like yield in $|\Delta\eta| < 0.78$ as a function of p_T^{trigger} for K_S^0 -h, and Λ -h correlations for 1.5 GeV/c $< p_T^{\text{associated}} < 3.0$ GeV/c in minimum bias d+Au, 0-60% Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. For comparison h-h correlations [7] from 40-80% Au+Au collisions are shown as a line. Peripheral Au+Au points have been shifted for visibility. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

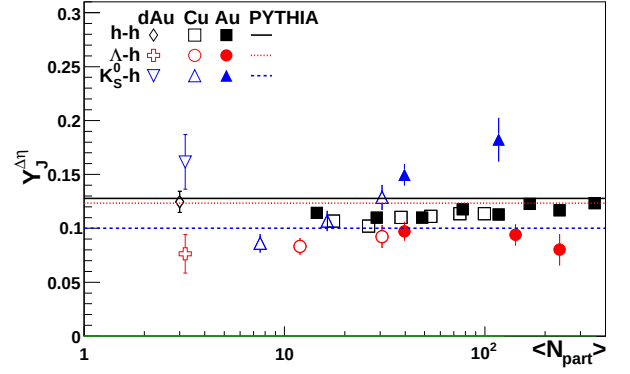


FIG. 4: Centrality dependence of the jet-like yield of K_S^0 -h, and Λ -h correlations for $3 < p_T^{\text{trigger}} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\text{associated}} < p_T^{\text{trigger}}$ in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data are compared to PYTHIA [40] calculations using the Perugia 2011 tune [41]. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

Λ -h jet-like yield may be lower in central Au+Au collisions, particularly at high p_T . For this reason we only compare to h-h correlations from 40-80% Au+Au collisions.

Next the jet-like yields are studied as a function of collision centrality expressed in terms of number of participating nucleons (N_{part}) calculated from the Glauber model [45]. The extracted jet-like yield as a function of N_{part} is shown in Figure 4 for h-h [7], K_S^0 -h, and Λ -h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. While there is no centrality dependence in the jet-like yield of h-h correlations, there is a centrality dependence in the yields of the K_S^0 -h and Λ -h correlations. These data are compared to PYTHIA [40] calculations from the Perugia 2011 [41] tune. There is a hint of a mass ordering, with the jet-like yield from K_S^0 -h correlations generally above that of the jet-like yield from h-h correlations and the jet-like yield from Λ -h generally below that of the h-h correlations. This is the opposite of the mass ordering observed in PYTHIA.

The jet-like yield as a function of $p_T^{\text{associated}}$ is shown in Figure 5 for K_S^0 -h, and Λ -h correlations for d+Au and Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. The Λ -h and K_S^0 -h correlations are only shown for d+Au and Cu+Cu collisions since residual track merging made measurements in Au+Au collisions difficult. Data are compared to the jet-like yield from h-h correlations [7]. The trend is similar for h-h, K_S^0 -h, and Λ -h correlations, however, there are hints that there may be some difference between systems.

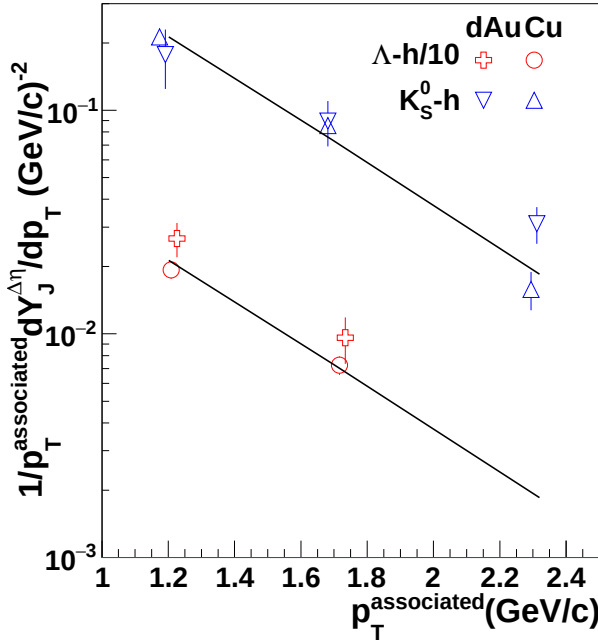


FIG. 5: The jet-like yield as a function of $p_T^{\text{associated}}$ for K_S^0 -h, and Λ -h correlations for $3 < p_T^{\text{trigger}} < 6$ GeV/c in d+Au, 0-60% Cu+Cu and 40-80% Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data are compared to the jet-like yield from h-h correlations [7], shown as a line. Data are binned in $1.0 < p_T^{\text{associated}} < 1.5$ GeV/c, $1.5 < p_T^{\text{associated}} < 2.0$ GeV/c, and $2.0 < p_T^{\text{associated}} < 3.0$ GeV/c and are plotted at the mean of the bin. The 5% systematic error due to the uncertainty on the associated particle's efficiency is not shown.

V. CONCLUSIONS

The $\frac{\Lambda+\bar{\Lambda}}{2K_S^0}$ ratio for the jet-like correlation is comparable to that observed in $p+p$ collisions, providing further evidence that this correlation is dominantly produced by fragmentation. In PYTHIA, this ratio is comparable to the inclusive ratio. We observe a trend for mass ordering of the yield of identified strange trigger particles in d+Au, Cu+Cu, and Au+Au collisions. This mass ordering is not the same ordering observed for the jet-like yield in PYTHIA.

These studies provide motivation for future studies of strangeness production in jets. Larger data sets and data from collisions at higher energies could provide more robust tests of the strangeness production mechanism in $p+p$ collisions. Studies in $p+p$ would be essential in order to search for modifications of strangeness production in jets in heavy ion collisions.

- [1] S. S. Adler et al. (PHENIX), Phys. Rev. Lett. **91**, 172301 (2003), nucl-ex/0305036.
- [2] J. Adams et al. (STAR), Phys. Rev. Lett. **92**, 052302 (2004), nucl-ex/0306007.
- [3] S. Chatrchyan et al. (CMS Collaboration), Eur.Phys.J. **C72**, 1945 (2012), 1202.2554.
- [4] K. Aamodt et al. (ALICE Collaboration), Phys.Lett. **B696**, 30 (2011), 1012.1004.
- [5] A. Adare et al. (PHENIX Collaboration), Phys.Rev.Lett. **98**, 162301 (2007), nucl-ex/0608033.
- [6] G. Agakishiev et al. (STAR Collaboration), Phys.Rev. **C86**, 014904 (2012), 1111.5637.
- [7] G. Agakishiev et al. (STAR Collaboration), Phys.Rev. **C85**, 014903 (2012), 1110.5800.
- [8] J. Adams et al. (STAR), Phys. Rev. Lett. **95**, 152301 (2005).
- [9] B. I. Abelev et al. (STAR), Phys. Rev. Lett. **102**, 052302 (2009).
- [10] C. Adler et al. (STAR), Phys. Rev. Lett. **90**, 082302 (2003), nucl-ex/0210033.
- [11] J. Adams et al. (STAR), Phys. Rev. Lett. **97**, 162301 (2006).
- [12] B. I. Abelev et al. (STAR), Phys. Rev. C **80**, 064912 (2009).
- [13] B. I. Abelev et al. (STAR), Phys. Rev. Lett. **105**, 022301 (2010).
- [14] H. Agakishiev et al. (STAR) (2010), nucl-ex/1010.0690.
- [15] K. Aamodt et al. (ALICE Collaboration), Phys.Rev.Lett. **108**, 092301 (2012), 1110.0121.
- [16] S. Chatrchyan et al. (CMS Collaboration), JHEP **1107**, 076 (2011), 1105.2438.
- [17] S. Chatrchyan et al. (CMS Collaboration), Eur.Phys.J. **C72**, 2012 (2012), 1201.3158.
- [18] G. Aad et al. (ATLAS Collaboration), Phys.Lett. **B719**, 220 (2013), 1208.1967.
- [19] G. Aad et al. (ATLAS Collaboration), Phys.Rev.Lett. **105**, 252303 (2010), 1011.6182.
- [20] S. Chatrchyan et al. (CMS Collaboration) (2013), 1310.0878.
- [21] B. Abelev et al. (ALICE Collaboration), JHEP **1203**, 053 (2012), 1201.2423.
- [22] A. Adare et al. (PHENIX), Phys.Rev. **C77**, 011901 (2008), 0705.3238.
- [23] A. Adare et al. (PHENIX), Phys.Rev.Lett. **104**, 252301 (2010), 1002.1077.
- [24] S. Adler et al. (PHENIX), Phys.Rev.Lett. **97**, 052301 (2006), nucl-ex/0507004.
- [25] J. Adams et al. (STAR) (0100), nucl-ex/0601042.
- [26] B. I. Abelev et al. (STAR), Phys. Rev. Lett. **97**, 152301 (2006), nucl-ex/0606003.

- [27] K. H. Ackermann et al. (STAR), Nucl. Instrum. Meth. **A499**, 624 (2003).
- [28] M. Anderson et al., Nucl. Instrum. Meth. **A499**, 659 (2003).
- [29] J. Adams et al. (STAR), Phys. Rev. Lett. **91**, 072304 (2003).
- [30] C. Adler et al. (STAR), Phys. Rev. Lett. **89**, 092301 (2002), nucl-ex/0203016.
- [31] M. Aggarwal et al. (STAR Collaboration), Phys.Rev. **C83**, 024901 (2011), 1010.0142.
- [32] G. Agakishiev et al. (STAR Collaboration), Phys.Rev.Lett. **108**, 072301 (2012), 1107.2955.
- [33] B. B. Back et al. (PHOBOS), Phys. Rev. C **72**, 051901(R) (2005).
- [34] B. B. Back et al. (PHOBOS), Phys. Rev. Lett. **94**, 122303 (2005).
- [35] B. Alver et al. (PHOBOS), Phys. Rev. Lett. **104**, 062301 (2010).
- [36] L. Adamczyk et al. (STAR), Phys.Rev. **C88**, 014904 (2013), 1301.2187.
- [37] B. Abelev et al. (STAR Collaboration), Phys.Lett. **B683**, 123 (2010), 0904.1722.
- [38] B. Abelev et al. (STAR Collaboration), Phys.Rev. **C75**, 064901 (2007), nucl-ex/0607033.
- [39] K. Aamodt et al. (ALICE Collaboration), Eur.Phys.J. **C71**, 1594 (2011), 1012.3257.
- [40] T. Sjostrand, S. Mrenna, and P. Z. Skands, JHEP **0605**, 026 (2006), hep-ph/0603175.
- [41] P. Z. Skands, Phys.Rev. **D82**, 074018 (2010), 1005.3457.
- [42] R. Field and R. C. Group (CDF) (2005), hep-ph/0510198.
- [43] K. Aamodt et al. (ALICE Collaboration), Phys.Lett. **B693**, 53 (2010), 1007.0719.
- [44] B. Abelev et al. (ALICE Collaboration), Phys.Lett. **B710**, 557 (2012), 1112.2222.
- [45] M. L. Miller, K. Reygers, S. J. Sanders, and P. Steinberg, Ann. Rev. Nucl. Part. Sci. **57**, 205 (2007), nucl-ex/0701025.